

Ordinary Differential Equations

MATH 2314 Spring 2027

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Last Updated: October 6, 2026

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Course Documents

Homework 01

Instructions: Complete all the exercises below and submit your work by the due date.

1. This is the first homework exercise.

Module FO: First-Order Differential Equations

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

FO1: Solutions to Differential Equations

Objectives

- I can determine if a function or family of functions is a solution to a given differential equation.

What Is a Differential Equation?

Definition 1 (Ordinary/Partial) Differential Equation. A **differential equation** is an equation which relates a function to its derivatives. There are two kinds of differential equations: ordinary differential equations (ODEs) and partial differential equations (PDEs). In this course, we will restrict our focus to ordinary differential equations. $\diamond$

Example 2 An example of an ordinary differential equation is the familiar

which asks the question: "What function $y(x)$ has a derivative equal to $2x$?" The solution to this differential equation is the family of functions

where C is an arbitrary constant. We can check that this is a solution by differentiating; $\frac{d}{dx}(x^2 + C) = 2x$, so we have satisfied the condition stated by the differential equation. $\square$

Example 3 Other examples of ordinary differential equations are

$\square$

Example 4 Below are examples of partial differential equations:

The wave equation:

The heat equation:

□

Testing Solutions to ODEs

Definition 5 Solution to a Differential Equation. A solution to a differential equation is a function which satisfies the rules imposed by the equation. ◇

Example 6 The family of functions $y(x) = x^2 + C$ are solutions to the differential equation

because $y'(x) = \frac{dy}{dx} = 2x$ □

Example 7 Exponential Growth. The family $x(t) = Ce^{kt}$ is a solution to the differential equation

for some constant $a \in \mathbb{R}$ because $x'(t) = Cke^{kt}$ and $x''(t) = Ck^2e^{kt}$. Setting $a = k^2$, we have $x''(t) = aCe^{kt} = ax(t)$.

This differential equation models *exponential growth*. In reality, few things grow exponentially due to constraints in the environment. Despite its lack of realism, we can use the exponential growth equation to help build other, more useful models. □

Example 8 Logistic Growth. A population which needs to compete for resources will not grow exponentially-- think of environmental constraints such as access to food, water, shelter. In the absence of predators, this population will grow up to a certain point and stabilize; we can modify the exponential growth equation (see [Example 7](#)) to □

Checkpoint 9 Show that the family $x(t) = C \cosh(kt) = \frac{1}{2}(e^t + e^{-t})$ is also a solution to the exponential growth equation.

Checkpoint 10 For the differential equation

Show that the families $f_1(x) = C_1e^{3x}$ and $f_2(x) = C_2xe^{3x}$ are both solutions to the ODE. What do you think might be true of $f_3(x) = f_1(x) + f_2(x)$?

FO2: Separation of Variables

test

FO3: The Integrating Factor

test

FO4: The Laplace Transform (First-Order Equations)

test

FO5: Applications of First-Order Differential Equations

test

FO6: Slope Fields

test

FO7: Equilibrium Points & Autonomous Equations

test

FO8: Bifurcations

test

Module SO: Second-Order Differential Equations

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

SO1: Linear Homogeneous Differential Equations

Objectives

- I can solve linear, homogenous differential equations of any order.

SO2: Linear Non-Homogeneous Differential Equations

Objectives

- I can solve linear, non-homogeneous differential equations of any order.

SO3: The Laplace Transform (Second-Order Equations)

Objectives

- I can use the Laplace transform to solve second-order, linear differential equations.

SO4: The Laplace Transform (Unit Step and Impulse Functions)

Objectives

- I can use the Laplace transform to solve differential equations involving unit step or impulse functions.

SO5: The Laplace Transform (Convolutions)

Objectives

- I can use the Laplace transform to compute the convolution of two functions.

SO6: Applications of Second-Order Differential Equations

Objectives

- I can solve applied problems resulting from second-order differential equations.

Module SD: Systems of Differential Equations

The following notes are intended to remind you what we covered each week. They are not a substitute for attending class and not a substitute for reading the textbook.

SD1: Solutions to Differential Equations

Objectives

- I can use matrix algebra to compute linear combinations, products of matrices, and determinants of matrices, as well as determine when these are not possible.

SD2: Eigenvalues & Eigenvectors

Objectives

- I can find the eigenvalues and eigenvectors for a 2×2 matrix.

SD3: Linear Independence

Objectives

- I can determine if two vectors are linearly independent.

SD4: Complex & Real Eigenvalues

Objectives

- I can solve a system of differential equations with complex or distinct real eigenvalues.

SD5: Repeated Eigenvalues

Objectives

- I can solve a system of differential equations with repeated real eigenvalues.

SD6: Applications of Systems of Differential Equations

Objectives

- I can use systems of differential equations to solve applied problems.